# NAFASS: Explainable Multi-Horizon Air-Quality Forecasting with Auditable Local-Worker Task Execution in Southern Tunisia

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**Abstract**

Air pollution in North Africa reflects the interaction of mineral dust, human activities and meteorological variability. In southern Tunisia, the proximity of urban, industrial and coastal environments motivates forecasting methods that distinguish short-term persistence from day-ahead variability. This study evaluates seven machine-learning and deep-learning configurations for service-derived air-quality index (AQI) forecasting in four zones of Gabès at lead times of 1, 6, and 24 hours. Tree ensembles, recurrent networks, attention, and autoencoder-based configurations use a common 54-feature schema and chronological training, validation, and test partitions. The archived test export contains 405,300 predictions covering 84 model–zone–horizon combinations. XGBoost, selected by mean validation RMSE, has four-zone mean test RMSEs of 0.515, 3.294, and 10.909 AQI points, compared with persistence errors of 1.067, 5.129, and 12.676. These descriptive differences correspond to approximate reductions of 51.7%, 35.8%, and 13.9%. Architectural complexity does not consistently improve the archived errors. At 24 hours, direct threshold recall is only 38.9% and 37.1% in the two highest diagnostic bands, despite useful average regression performance. The project also implements a multi-method explanation layer: TreeSHAP and SHAP beeswarms, permutation importance, partial-dependence plots and a custom LIME-style local surrogate for tabular models, together with GradientExplainer summaries for sequence models. Exported XAI artifacts cover all 84 model–zone–horizon units. The implementation, split labels, feature counts and stored values are verified, but the artifact metadata do not contain immutable checkpoint hashes or a complete input-manifest link; the explanations are therefore presented as audit-linked descriptive model behaviour, not causal or fully reproducible findings. A zone-wise worker layer records task-level execution through independent local processes. The worker run completed 84 tasks after retries; its unmatched test populations do not support a comparison of predictive accuracy or speed. The findings therefore support a horizon-specific assessment of the archived AQI series, not a validated operational warning system. Unresolved acquisition timing, historical preprocessing linkage, and the absence of independent station validation limit conclusions about operational forecasting and local pollution exposure.

**Keywords:** air-quality forecasting; explainable artificial intelligence; TreeSHAP; LIME; machine learning; deep learning; temporal validation; rare pollution episodes; North Africa; southern Tunisia.

# 1 Introduction

Air pollution in North Africa presents a forecasting problem in which natural dust transport and human activities interact with changing meteorological conditions. The Sahara is a major source of mineral dust transported across the Mediterranean, while urban and industrial environments introduce additional pressures. Elevated particulate levels therefore cannot be interpreted solely through one emission source or one temporal scale. Forecasting requires models that capture recent persistence without assuming that the same relationships remain adequate at longer lead times. The World Health Organization's guidelines establish the health relevance of the main ambient pollutants [1]; this study addresses prediction rather than estimating exposure-related health outcomes.

Tunisia lies at the intersection of Mediterranean atmospheric circulation and Saharan influence. Measurements from Tunisian sites have documented dust-related increases in particulate concentrations and the influence of ventilation, precipitation and seasonal circulation [2]. This motivates the inclusion of pollutant histories, dust and meteorological variables in forecasting models. It does not imply that the causal effect of each predictor can be identified from a prediction benchmark.

The southern Tunisian setting is particularly relevant to the present application. In Gabès, on the southeastern coast, urban areas, industrial activities and coastal conditions occur in close proximity. A local study documented spatial and temporal variation in deposited particles associated with source environments and meteorology [3]. Particle deposition is nevertheless different from the atmospheric AQI series analysed here. Gabès Ville, Ghannouche, Chott Salem and Teboulbou are treated as application zones rather than independently authenticated monitoring stations or industrial-source attribution sites.

Artificial intelligence (AI) provides several ways to represent temporal environmental relationships. Tree ensembles learn nonlinear interactions among engineered variables, whereas recurrent networks, attention and autoencoders learn representations from historical sequences. The scientific question is whether these additional representation mechanisms improve forecasts under a common evaluation design. Published air-quality studies use diverse targets, locations and protocols [4–6]; concentration error, AQI error and category-classification accuracy are not interchangeable.

This study reframes NAFASS as a multi-horizon AI benchmark. It asks: (i) how do tree-based and sequence-based configurations compare at 1, 6 and 24 hours; (ii) does validation-selected forecasting improve on aligned persistence; and (iii) does low average error translate into recognition of high-AQI observations? It also examines which conclusions remain defensible when upstream data provenance and historical preprocessing linkage are incomplete.

The contribution is an application-oriented comparison rather than a new forecasting algorithm. It combines seven configurations, four zones and three horizons; evaluates errors against matched simple references; and separates aggregate regression performance from rare-band recognition. Validation evidence determines selection before test reporting. The computational layer also records task-level execution: a coordinator allocates independent model–horizon tasks to zone-specific workers. The inspected implementation runs concurrent local processes; it is not evidence of multi-node execution, distributed gradient training, or a measured speedup. The XAI layer is a genuine implemented component, but its exported explanations are reported with an explicit checkpoint-linkage qualification: they support audit-linked descriptive interpretation, not causal attribution or full end-to-end reproducibility.

# 2 Related work and scientific positioning

## 2.1 AI approaches to air-quality forecasting

Fine-grained air-quality prediction combines heterogeneous environmental information and temporal structure [4, 5]. Tree ensembles are useful references for structured data containing AQI lags, pollutant summaries and weather variables. XGBoost implements regularised boosted trees [7]; its relevance here lies in representing nonlinear interactions, not in a new boosting procedure introduced by this study.

Sequence models supply different inductive biases. LSTM and bidirectional LSTM configurations encode a historical window; self-attention connects positions within a sequence [8]; and autoencoders learn representations through reconstruction. None guarantees a forecast improvement. Fair assessment requires attention to common information windows, tuning budgets, optimisation, preprocessing and model selection. A ranking alone cannot identify the causal benefit of an architectural component.

Modern sequence-forecasting research also motivates stronger simple references and additional comparators. Linear forecasting [9], temporal convolutional networks [10] and patch-based transformers [11] are relevant to a future expanded benchmark. These models were not evaluated in the archived experiments and are not presented as completed results.

## 2.2 Recent advances and the remaining local forecasting gap

Recent work extends the forecasting landscape beyond recurrent hybrids. Aurora demonstrates a pretrained Earth-system representation that can be adapted to air-quality tasks [19]. It motivates testing transferable representations, but its global gridded task differs from predicting a service-derived AQI at four nearby locations. A smaller local benchmark cannot claim comparable scope or numerical superiority.

X2-AQFormer couples multi-day pollutant prediction with an interpretable transformer framework [20]. Its use of monitoring-site data and dynamic input attribution highlights the value of linking explanations to a defined forecasting task. SynCast combines a regional transformer with stochastic refinement to improve the representation of particulate extremes [21]. These studies motivate stronger temporal models and tail-sensitive evaluation, rather than adding architectural complexity without controlled evidence.

Fahim and Karim's recent cost-aware evaluation compares specialised and foundation-model forecasting across cities while accounting for covariate timing [22]. It reinforces the need to distinguish deployable inputs from perfect future information. A recent review further organises the roles of non-neural, deep and hybrid approaches and identifies generalisation and interpretability as continuing challenges [23]. For NAFASS, the resulting gap is a carefully controlled, locally grounded comparison with explicit temporal validity and rare-observation diagnostics. None of the newer architectures is claimed to have been trained in the present experiments.

## 2.3 Temporal validation and explainability

Environmental series are dependent in time and sometimes across locations. Validation must respect these structures [12]. Chronological splitting is necessary for forecasting, but does not prevent leakage if future records enter preprocessing or predictors were unavailable when a forecast would have been issued. Such problems can undermine apparently strong ML results [13].

Explanations address model behaviour rather than environmental causation. SHAP attributes a model output relative to a reference [14]; it cannot alone identify pollution sources. An attention matrix is likewise an internal model quantity rather than a validated map of atmospheric influence.

An explanation should remain linked to its checkpoint, feature order, reference sample and explained observations.

Rare high-AQI observations demand separate evaluation. A model can perform well under average RMSE while missing infrequent high-index transitions. Accordingly, this study retains band supports and raw-threshold confusion matrices. It does not combine undocumented classification transformations with thresholded continuous outputs as if they were one verified metric pipeline.

# 3 Study area, data and forecasting task

## 3.1 Four application zones in Gabès

Models are fitted separately for Z1 (Gabès Ville), Z2 (Ghannouche), Z3 (Chott Salem) and Z4 (Teboulbou). These units enable zone-specific reporting. They do not establish transfer to unseen cities, and four nearby application zones should not be interpreted as four independent external datasets. The North African context motivates the problem; the empirical scope remains the four Gabès series.

## 3.2 Service-derived data and acquisition uncertainty

The original scripts request Open-Meteo air-quality variables, including `us_aqi`, particulate matter, gaseous pollutants and dust. Forecast and historical-forecast services provide weather enrichment. Open-Meteo documents CAMS-based gridded products and an AQI consolidated from pollutant subindices [16]. The series must therefore be described as service-derived, not independently verified station measurements. The CAMS reanalysis literature provides product context [17], but does not authenticate the exact product vintage of each application record.

The original acquisition description records a `models=cams_europe` request, while current air-quality documentation uses `domains` for domain selection. Historical request/response payloads, the effective domain, resolved grid cells, release times and product version must be recovered before attributing records to a particular source product. The distinction matters when nearby coordinates may resolve to the same or neighbouring grid cells. No claim of neighbourhood-scale atmospheric resolution is established by named zones alone.

The archived prediction export has six fields: zone, target timestamp, horizon, model, reference AQI and predicted AQI. It contains 405,300 finite records for 84 combinations with 4,825 pairs each. Target times run from 14 March 2026 at 20:00 to 1 October 2026 at 20:00. The export's timezone is unspecified; UTC acquisition metadata does not retrospectively certify it.

A later database contains 24,660 hourly rows per study zone, from 1 January 2024 to 24 October 2026 at 11:00. The supplied audit reports exact agreement between its AQI values and the 19,300 distinct zone–target-time observations in the export. This supports internal target consistency, but the later database does not replace the original frozen training snapshot. Complete acquisition history and immutable input-to-run linkage remain unavailable.

## 3.3 Target and information set

For zone $z$, forecast origin $t$ and lead time $h \in \{1, 6, 24\}$ hours,

$$y_{z,t,h} = \mathrm{AQI}_{z,t+h}, \qquad \widehat{y}_{z,t,h} = f_{z,h}(\mathcal{I}_{z,t}). \tag{1}$$

The information set $\mathcal{I}_{z,t}$ should contain only quantities available by issue time. Tree models use a vector of 54 features. Neural models use a historical tensor of shape $L \times 54$, with $L = 24$ for 1- and 6-hour prediction and $L = 168$ for 24-hour prediction. Target-calendar values are known in advance; realised future weather and AQI are not.

Timestamp indexing and operational availability are distinct. A retrospectively enriched weather record or profile aggregate can be indexed at $t$ without having been available at issue time. Predictor release and revision histories therefore require verification independently of the array indexing.

### 3.4 Common feature representation

Table 1 summarises the 54-feature schema. The same three fuzzy-context inputs are included in the common schema used by all seven configurations. This revision therefore calls the boosted-tree model *XGBoost*: the original label "XGBoost + Fuzzy" did not designate a fuzzy forecasting component exclusive to that model.

**Table 1: Shared predictor families. Historical support ends at the forecast origin; target-calendar variables are known in advance.**

| Family | Count | Main content |
|---|---|---|
| AQI history | 10 | Current AQI; lags 1–7, 24 and 168 hours |
| AQI summaries | 5 | Differences, trailing means and variability |
| Fuzzy context | 3 | Context score and lower/upper assessment bounds |
| Current pollutants | 7 | $PM_{2.5}$, $PM_{10}$, $SO_2$, $NO_2$, $O_3$, CO and dust |
| Current weather | 7 | Temperature, relative humidity, wind speed and direction, surface pressure, precipitation, and cloud cover |
| Calendar | 5 | Target hour, weekend, season and cyclic hour encoding |
| Pollutant history | 7 | Lags, particulate summaries, slope and ratio |
| Wind transforms | 3 | Two code-defined components and a calm indicator |
| Rain summaries | 2 | Trailing accumulation and time since rain |
| Weather enrichment | 5 | Dew point, vapour deficit, gusts, low cloud and 80-m wind |
| Total | 54 | Common names; tabular and sequence representations differ |

Fuzzy context summarises zone-level profiles and recorded application events. Its bounds are assessment outputs, not calibrated forecast intervals. Timestamp filters alone do not certify historical profile versioning, consent or operational availability. A future ablation should compare the full schema with an environmental-only version excluding those three inputs. Their incremental value is not established by the present ranking.

# 4 Models and experimental protocol

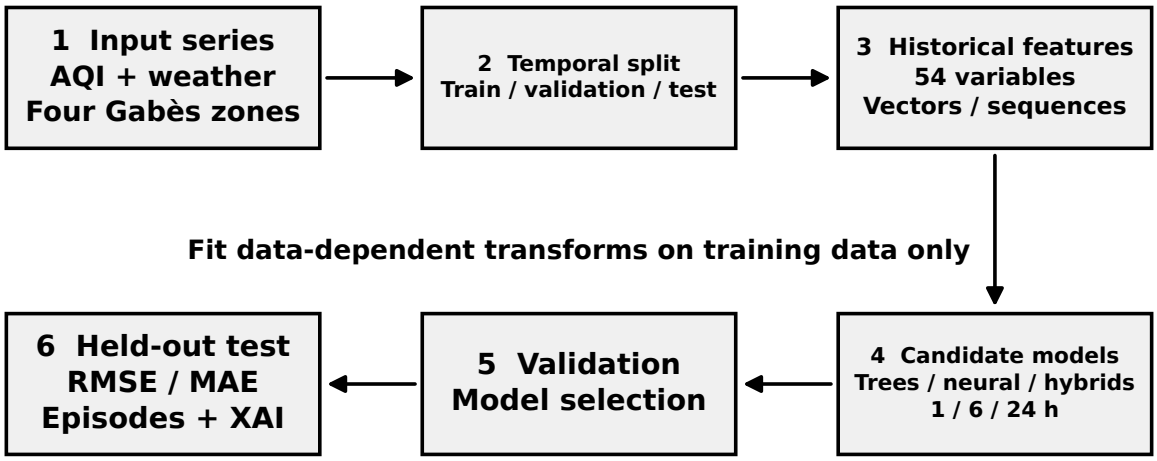



**Figure 1: Interconnected forecasting workflow. The diagram summarises the reference design; historical preprocessing and data-availability uncertainties are detailed in the text.**

## 4.1 Tree ensembles

Random Forest and XGBoost use the common tabular representation. Recorded Random Forest settings specify 200, 300 and 400 trees for the three horizons, with maximum depths of 12, 10 and 8. Minimum split sizes are 10, 12 and 20, and minimum leaf sizes are 2, 4 and 8. The exported seed is 42.

The recorded XGBoost Optuna search [18] comprises 20 trials per zone at 1 hour, 30 at 6 hours and 40 at 24 hours, totalling 360 trials. The documented domain includes 200–900 estimators, learning rates of 0.01–0.12, and horizon-dependent depth constraints. The five overlapping searched parameters agree between trial minima and saved summaries. Other regularisation settings occur only in trial records; complete checkpoint configurations are still required for reproduction.

The inspected implementation permits validation-derived affine regression calibration. Its acceptance rules include a slope in $[0.5, 2.5]$, at least 3% validation-RMSE improvement, a strict absolute-error reduction for at least half the validation observations, and no decline in an internal class score. A separate dispersion transform can affect classification. These are documented options, not evidence that every archived unit used a non-identity correction or that the reported export used the same transform for every unit. Realised transforms must be recorded and must not be estimated using test data.

## 4.2 LSTM, BiLSTM and attention

The recurrent baselines begin with causal Conv1D (32 filters, kernel 3) and batch normalisation. The unidirectional LSTM uses recurrent layers with 64 and 32 units. In the BiLSTM and attention–BiLSTM configurations, these are implemented as Bidirectional(LSTM(64)) and Bidirectional(LSTM(32)) with the default concatenation merge: the units are 64 and 32 per direction, producing 128- and 64-dimensional outputs, respectively. All recurrent variants use dropout 0.2, a 32-unit ReLU dense layer and a scalar output. Their names describe the principal temporal

component rather than the absence of convolutional preprocessing.

The attention variant applies four-head self-attention with key dimension 16 after the first BiLSTM sequence and dropout. Attention output is added to the sequence and layer-normalised before the final recurrent block. All positions belong to the historical input window. Bidirectional processing of that historical window does not itself introduce samples after the forecast origin.

The recurrent module specifies Huber loss, Adam at $10^{-3}$, gradient clipping norm 1.0, batch size 64 and at most 60 epochs. Selection uses patience 8 for early stopping and a learning-rate reduction factor of 0.5 with patience 4. The selected epoch count determines a fresh train-plus-validation fit. These defaults differ from the realised epochs of individual archived checkpoints.

### 4.3 Autoencoder configurations

The LSTM autoencoder has an encoder with 64 and 16 units. A repeated 16-dimensional latent vector is decoded through 16- and 64-unit recurrent layers to reconstruct the 54 features at each step under mean squared error. The encoder is then frozen. A separate BiLSTM processes the raw window; its representation is concatenated with the latent vector before the forecast head. Thus, "BiLSTM–AE" denotes a two-stage forecasting system, not a bidirectional reconstruction encoder. The module's default maxima are 30 reconstruction epochs and 40 forecasting epochs, with early-stopping patience 8 during selection.

The CNN–AE has causal convolutions with 64 filters and kernel 5, followed by 32 filters and kernel 3, and a 16-channel latent projection. One branch reconstructs the input. The forecast branch concatenates the last latent step, average and maximum latent pooling, and the last original input vector. Dense layers of 64 and 32 units precede scalar output. Its joint objective is

$$\mathcal{L} = \mathcal{L}_{\text{Huber}}(y, \widehat{y}) + 0.0005\, \mathcal{L}_{\text{MSE}}(X, \widehat{X}). \tag{2}$$

The shortcut retains recent input information while reconstruction supplies an auxiliary objective. Their individual benefits require controlled ablations.

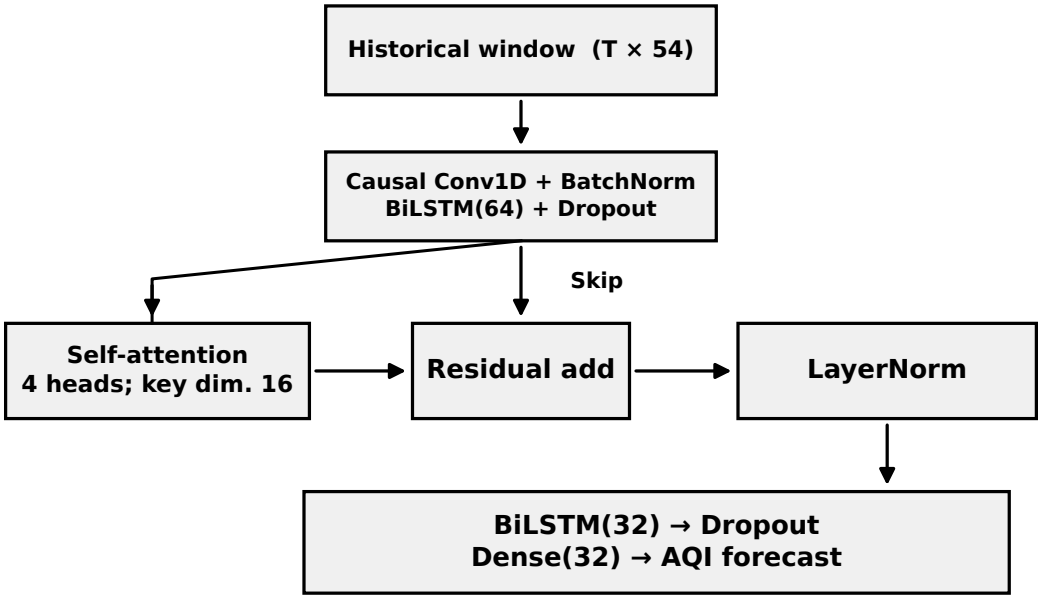



**Figure 2: Attention–BiLSTM forecasting path. Self-attention and the residual skip use the same historical sequence.**

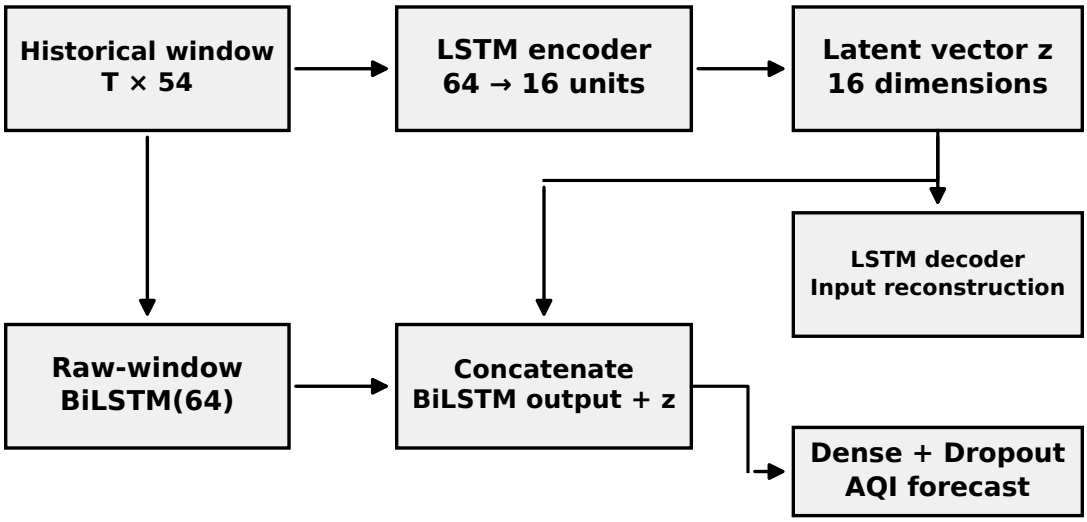



**Figure 3: Two-stage BiLSTM–AE. Reconstruction pretraining precedes a forecast fit using the frozen latent representation and a separate raw-window BiLSTM.**

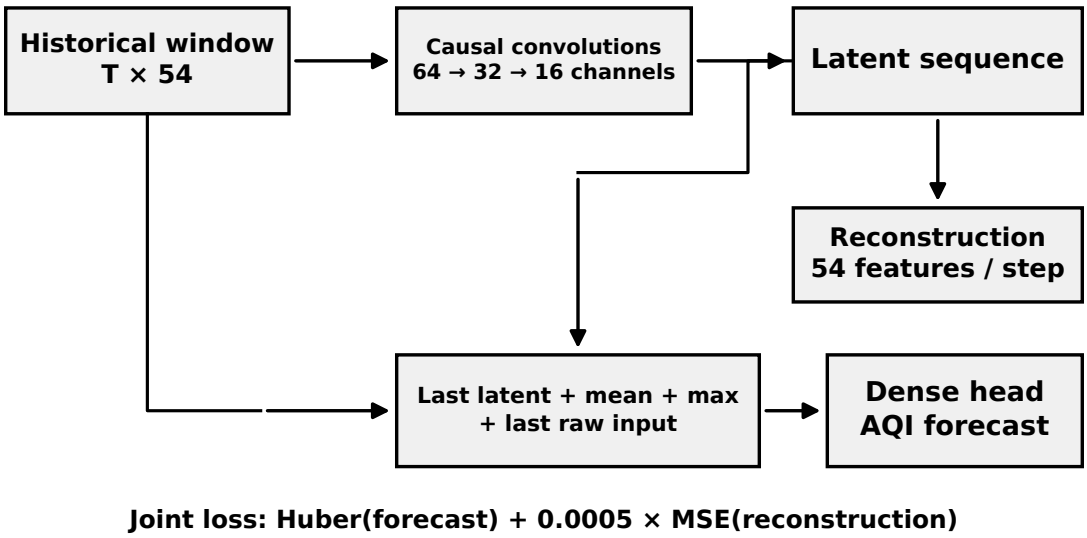



**Figure 4: Joint CNN–AE. The latent sequence feeds reconstruction and forecasting; the forecast also receives the last raw input.**

### 4.4 Temporal partitioning, preprocessing and selection

The archived protocol assigns 70% of the ordered series to training, 10% to validation, and 20% to testing. Recorded Z1 boundaries are 1 January 2024–4 December 2025 at 08:00 for training; 4 December 2025 at 09:00–14 March 2026 at 19:00 for validation; and 14 March 2026 at 20:00–1 October 2026 at 20:00 for testing. Log boundaries do not replace a complete sample-membership manifest.

The supplied source materials describe a corrected loader that forward-fills predictors only, with a maximum of three hourly records, and excludes missing AQI targets. An archived source snapshot also contains a pre-split interpolation path. Because the historical run is not immutably linked

to a specific loader revision and input snapshot, its metrics cannot be certified as outputs of the corrected loader; they are reported as archived results. The later four-zone snapshot reportedly has no numeric missingness, but that does not establish historical missingness. Results are retained as retrospective archived results rather than relabelled as a leakage-free rerun.

The historical protocol also lacks an established strict horizon-width purge. In a prospective-valid rerun, fitted targets must be available before the next evaluation issue period, with the same rule applied at the validation–test boundary. Scaling, imputation, representation learning and all data-dependent transformations must be fitted within the appropriate training partition.

The reported global diagnostic selection uses mean validation RMSE across the four zones, separately for each horizon. A separate deployment registry makes zone-specific choices. XGBoost is globally selected at all three horizons. The zone registry selects CNN–AE for Z1 at 6 hours and XGBoost for the other eleven zone–horizon combinations. These rules remain distinct; the best test error does not retrospectively change selection.

### 4.5 Weights, reference forecasts and metrics

The 24-hour forecast code supplies bounded class-frequency weights:

$$w_i = \mathrm{clip}\left(\sqrt{\frac{N}{4\max(n_{c(i)},1)}}, 0.75, 3\right), \tag{3}$$

where $N$ and $n_c$ refer to the fitted or monitored partition. Final fitting uses train-plus-validation frequencies. Reconstruction has its own loss weights. The benefit of this weighting is not established without an unweighted comparison.

Persistence predicts $\widehat{y}_{z,t,h} = \mathrm{AQI}_{z,t}$. The moving-average reference uses the 24 observations ending at $t$. Both references align with the archived forecast targets. For matched test pairs,

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (\widehat{y}_i - y_i)^2}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_i |\widehat{y}_i - y_i|. \tag{4}$$

Tables report arithmetic means of zone-specific errors, not pooled RMSE. Baseline-relative reduction is $100(1 - \overline{\mathrm{RMSE}}_m/\overline{\mathrm{RMSE}}_{\mathrm{pers}})$: a ratio of means, not the mean of zone-specific percentage gains.

The project's diagnostic thresholds are retained but named B1 ($y < 50$), B2 ($50 \le y < 100$), B3 ($100 \le y < 150$) and B4 ($y \ge 150$). This avoids confusing custom labels with standard U.S. AQI categories. B4 is not the standard U.S. "hazardous" band. The exact inequalities support reproduction, not regulatory reclassification. Because the stored macro-F1 transformation is not fully recoverable from continuous predictions, raw-threshold matrices provide the primary classification evidence in this revision.

### 4.6 Multi-method explanation protocol and evidence limits

NAFASS contains a genuine multi-method XAI layer. The audit examined its implementation, the training integration, the exported artifact table and the saved model inventory; it did not retrain models or recompute attributions. The 54-feature order is shared by the tabular and sequence representations, while the neural input windows use $L = 24$ for +1 h and +6 h and $L = 168$ for +24 h.

For Random Forest and XGBoost, the pipeline uses `shap.TreeExplainer`. For input $x$,

$$f(x) = \phi_0 + \sum_{j=1}^{54} \phi_j(x), \qquad I_j = \frac{1}{n}\sum_i |\phi_j(x_i)|. \tag{5}$$

The preferred mode is interventional TreeSHAP with up to 200 reference observations sampled from the fitting partition and up to 300 explained observations from the held-out test partition. The recorded XGBoost artifacts use the documented `tree_path_dependent` fallback, whereas Random Forest records use interventional TreeSHAP. Global $I_j$ measures contribution magnitude, not direction. SHAP beeswarms retain signed contributions for sampled test observations; feature colours are descriptive within-feature scales.

The same tabular units additionally export permutation importance (five repeats, RMSE scoring, up to 800 test observations), partial-dependence curves (up to six selected features, 20 points between the fifth and ninety-fifth test percentiles), and a custom LIME-style local weighted-ridge surrogate. The latter follows the LIME procedure [15] but is not the external `lime.lime_tabular` package: it uses 1,000 perturbations around one test instance and reports a local surrogate-fidelity diagnostic. These methods answer different questions and must not be merged into one importance measure. The stored local TreeSHAP records retain only leading contributions, so exact full additivity cannot be independently verified from every exported local payload.

For sequence models, the implementation uses `shap.GradientExplainer`, not DeepSHAP, and targets the forecast branch when a model also has a reconstruction output. Feature and temporal summaries are obtained by aggregating absolute attributions across the sequence and across explained test samples. The artifact table contains 60 gradient-SHAP summaries and 60 temporal-importance summaries for the five neural configurations. Attention is treated as an internal model quantity, not as a physical explanation.

The exported records cover 288 `nafas-xai-v2` artifacts across all 84 model–zone–horizon units: 24 global TreeSHAP, 24 beeswarm, 24 local TreeSHAP, 24 permutation, 24 PDP, 24 custom LIME-style, and 24 context/test-unit records for the 24 Random-Forest/XGBoost units, plus 60 GradientExplainer feature summaries and 60 temporal-importance summaries for the 60 neural units. The context records store the latest test prediction and test-row count; they are registry metadata, not an additional explanation method. Implementation presence, test-split labels, feature counts, sample limits and finite stored values are verified. However, the XAI rows do not contain an immutable checkpoint hash, complete preprocessing manifest or exact input-array identity. Matching per-unit test metrics and the saved checkpoint inventory provide strong circumstantial support for run association, but not immutable checkpoint linkage. The results are therefore reported as audit-linked descriptive explanations. Correlated AQI, pollutant and weather features must not be treated as causal source attribution.

### 4.7 Independent local-worker task decomposition

The computational layer uses independent zone-worker execution for the 84 model–zone–horizon combinations (Figure 5). Each worker fits and evaluates its assigned tasks independently. The inspected implementation launches concurrent local subprocesses and uses filesystem-backed state rather than a message broker or shared-gradient learning. Recorded explanations complement inspection of the models; their identity with evaluated checkpoints remains subject to the provenance checks described above.

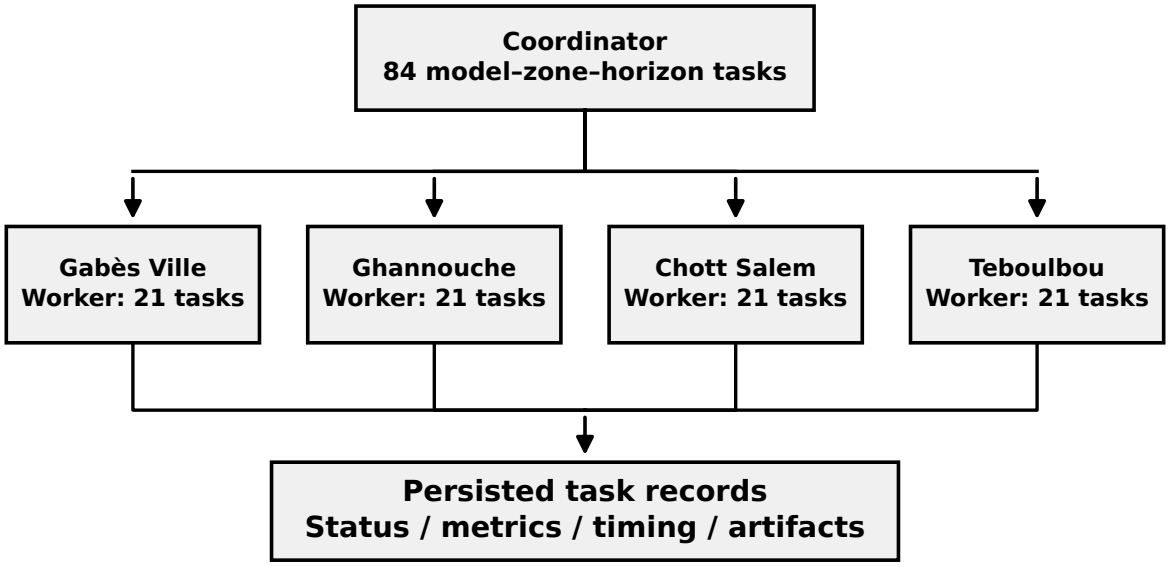



**Figure 5: Coordinator–worker execution principle. Each zone worker receives 21 independent tasks; persisted records support subsequent inspection. The inspected implementation is local-process based, without an established multi-node speedup.**

### 4.8 Pre-specified reporting rules

The following rules govern interpretation of the archived benchmark. (i) The information set ends at the forecast origin; future AQI and realised future weather are not admissible predictors. (ii) Scaling, imputation and other data-dependent transformations must be fitted within the relevant training partition. (iii) Global model selection uses mean validation RMSE by horizon; the held-out test set is used only after selection. (iv) Persistence predicts the AQI at the origin, while the moving reference is the mean of the 24 observations ending at that origin. (v) Diagnostic bands are B1 ($y < 50$), B2 ($50 \leq y < 100$), B3 ($100 \leq y < 150$) and B4 ($y \geq 150$); these are project thresholds, not a regulatory relabelling. (vi) Direct thresholding of continuous forecasts is reported separately from stored macro-F1 values whose transformation is not fully recoverable. (vii) The coordinator–worker layer records independent local task execution; it is not evidence of a multi-node cluster or speedup. (viii) All numerical claims are limited to the supplied service-derived export and are not presented as independent station validation.

## 5 Results

### 5.1 Model performance across horizons

The reported export covers $7 \times 4 \times 3 = 84$ combinations with 4,825 pairs each. The original audit reports agreement between recomputed and stored RMSE within 0.001 AQI. Its 405,300 rows represent repeated model predictions of 19,300 distinct zone–target-time values, not that many independent environmental observations.

Table 2 and Figure 6 show substantial differences between configurations. XGBoost has the lowest descriptive mean test RMSE at 1 and 24 hours; Random Forest has the lowest at 6 hours. The 6-hour validation criterion nevertheless selects XGBoost. This ranking difference is retained rather than using test scores for deployment selection.

**Table 2: Reported mean zone errors from the archived benchmark. All units contain 4,825 pairs. Bold identifies the lowest descriptive test RMSE, not a test-based selection rule.**

| | +1 h | | +6 h | | +24 h | |
|---|---|---|---|---|---|---|
| Model | RMSE | MAE | RMSE | MAE | RMSE | MAE |
| Random Forest | 2.743 | 1.967 | **2.373** | 1.142 | 11.045 | 5.569 |
| XGBoost | **0.515** | 0.275 | 3.294 | 1.777 | **10.909** | 5.897 |
| LSTM | 23.695 | 18.504 | 21.661 | 17.786 | 16.091 | 10.373 |
| BiLSTM | 17.009 | 13.386 | 12.819 | 10.083 | 18.327 | 13.289 |
| BiLSTM–MHA | 15.870 | 11.950 | 14.788 | 11.454 | 15.162 | 10.319 |
| BiLSTM–AE | 3.894 | 3.333 | 3.669 | 2.548 | 11.176 | 7.752 |
| CNN–AE | 1.888 | 1.373 | 3.356 | 2.231 | 20.265 | 15.388 |

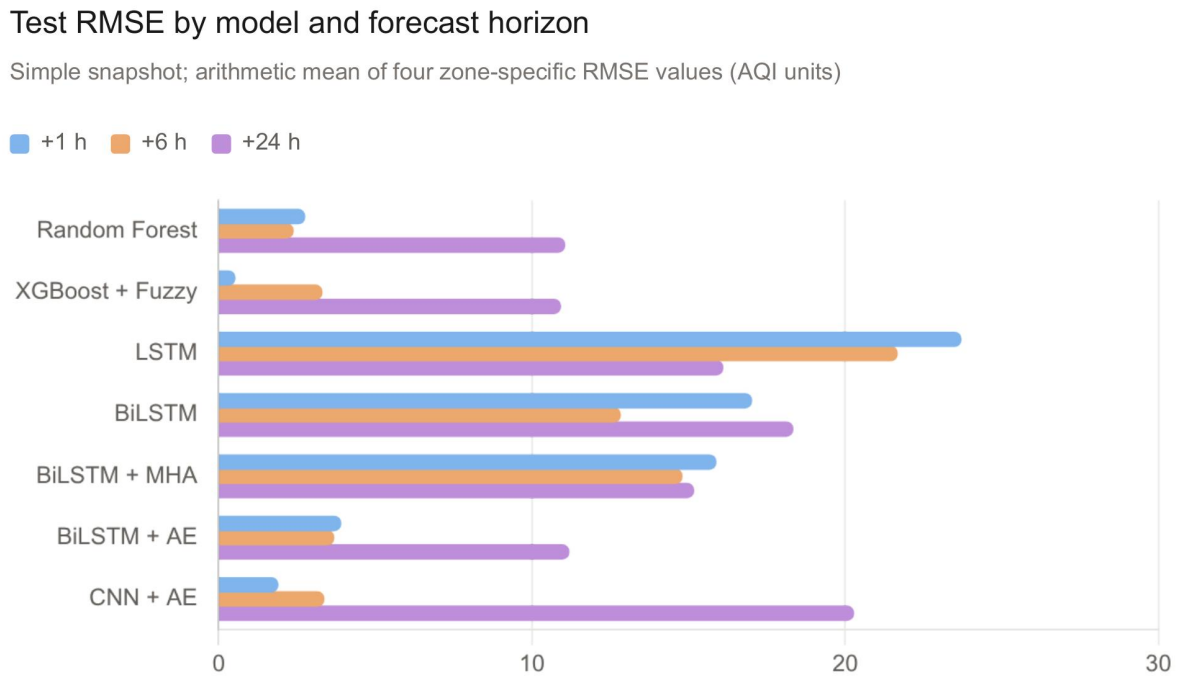



**Figure 6: Supplied benchmark RMSE figure, reproduced without redrawing. The source graphic retains the legacy label "XGBoost + Fuzzy"; in the manuscript this denotes XGBoost using the common feature schema. Bars retain the supplied values and scale.**

The neural results are heterogeneous. CNN–AE is comparatively effective at 1 hour but has a much larger 24-hour error. BiLSTM–AE is below the plain recurrent configurations at 1 and 6 hours and approaches the tree errors at 24 hours. These comparisons do not isolate reconstruction: architecture, training schedule and objectives change together. Attention also does not uniformly improve results across horizons. It is incorrect to assert that every model's error increases monotonically with lead time.

The large gap between some recurrent models and persistence warrants implementation checks before generalising about deep learning. Scaling and inverse transforms, target alignment, sequence construction, optimisation and learning curves should be inspected on identical frozen inputs. Poor performance from one configuration does not demonstrate an inherent weakness of a model family.

## 5.2 Validation-selected models against simple references

XGBoost is globally selected with validation RMSEs of 0.715, 5.278 and 18.444 at 1, 6 and 24 hours. Its test errors are 0.515, 3.294 and 10.909. Table 3 separates this validation choice from best-test rankings.

**Table 3: Aligned reference errors and validation-selected XGBoost. Percentage reductions use rounded mean RMSE values and are approximate.**

| Horizon | Persistence | 24-h moving mean | XGBoost | Reduction vs persistence |
|---|---|---|---|---|
| +1 h | 1.067 | 7.405 | 0.515 | 51.7% |
| +6 h | 5.129 | 9.508 | 3.294 | 35.8% |
| +24 h | 12.676 | 13.957 | 10.909 | 13.9% |

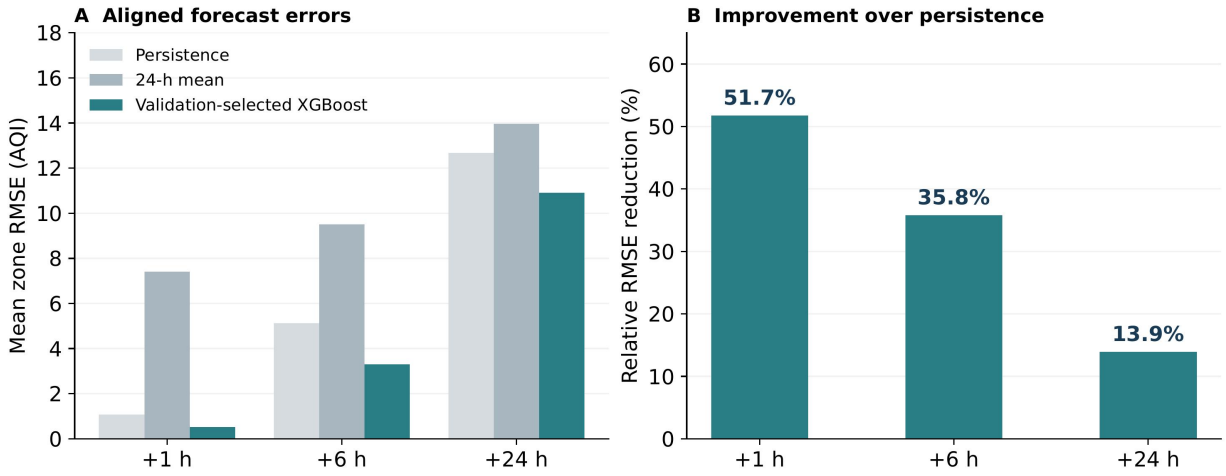



**Figure 7: Validation-selected XGBoost against aligned references (A), and relative RMSE reduction against persistence (B). Values are descriptive and use rounded four-zone means.**

The declining relative gain indicates that a short-horizon advantage need not persist at the same magnitude for day-ahead prediction. This is not a demonstrated atmospheric mechanism. A service-derived target can inherit smoothness and dependencies from its upstream model. Independent observations and original issue-time forecasts would help quantify added value beyond the atmospheric provider itself.

## 5.3 Full-period temporal behaviour across zones

Figures 8 and 9 reproduce the archived full-test-period curves for all four zones and three horizons. The globally validation-selected model is XGBoost in every panel; the historic model label is preserved where it appears inside the source graphics. Each panel retains 4,825 reference–prediction pairs. Common axes and a common period enable visual inspection without changing amplitudes or selecting a particularly favourable episode.

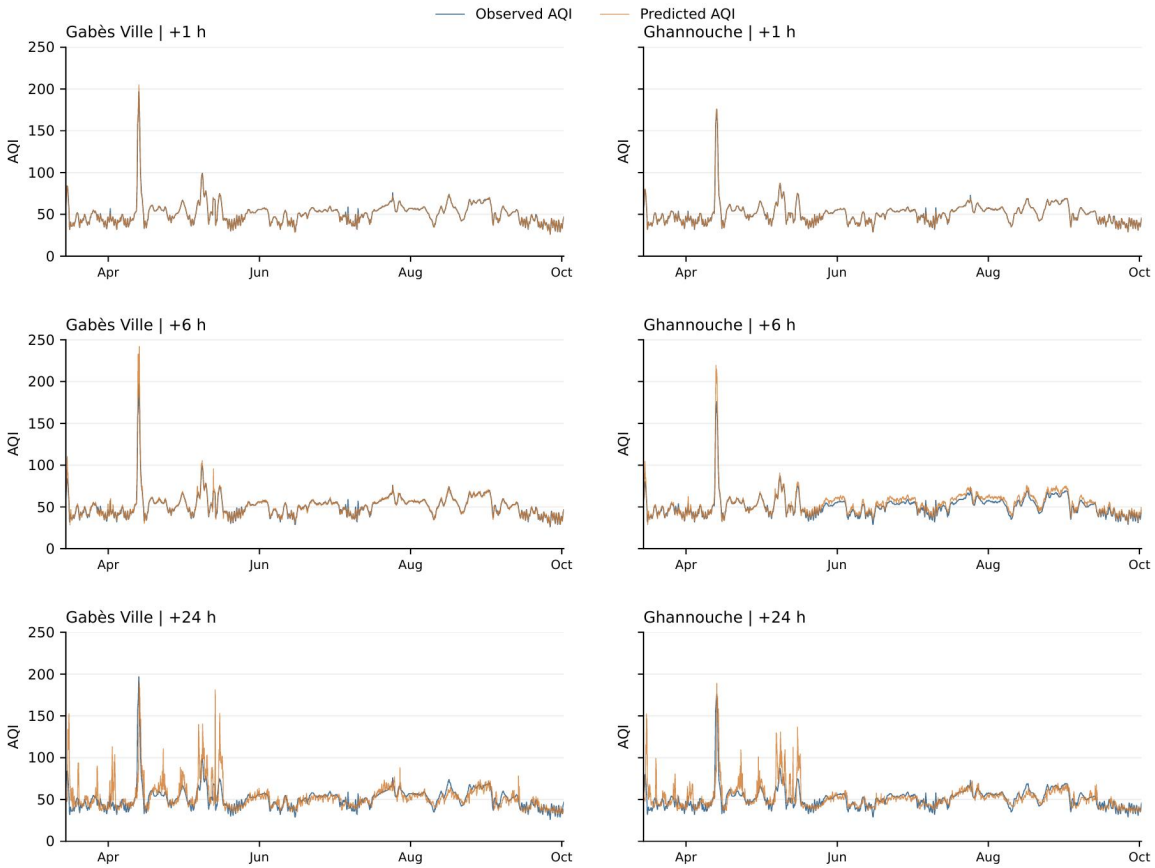



**Figure 8: Original full-period AQI curves for Gabès Ville and Ghannouche at 1, 6 and 24 hours. XGBoost is globally validation-selected. Curves and axes are preserved; original colour presentation.**

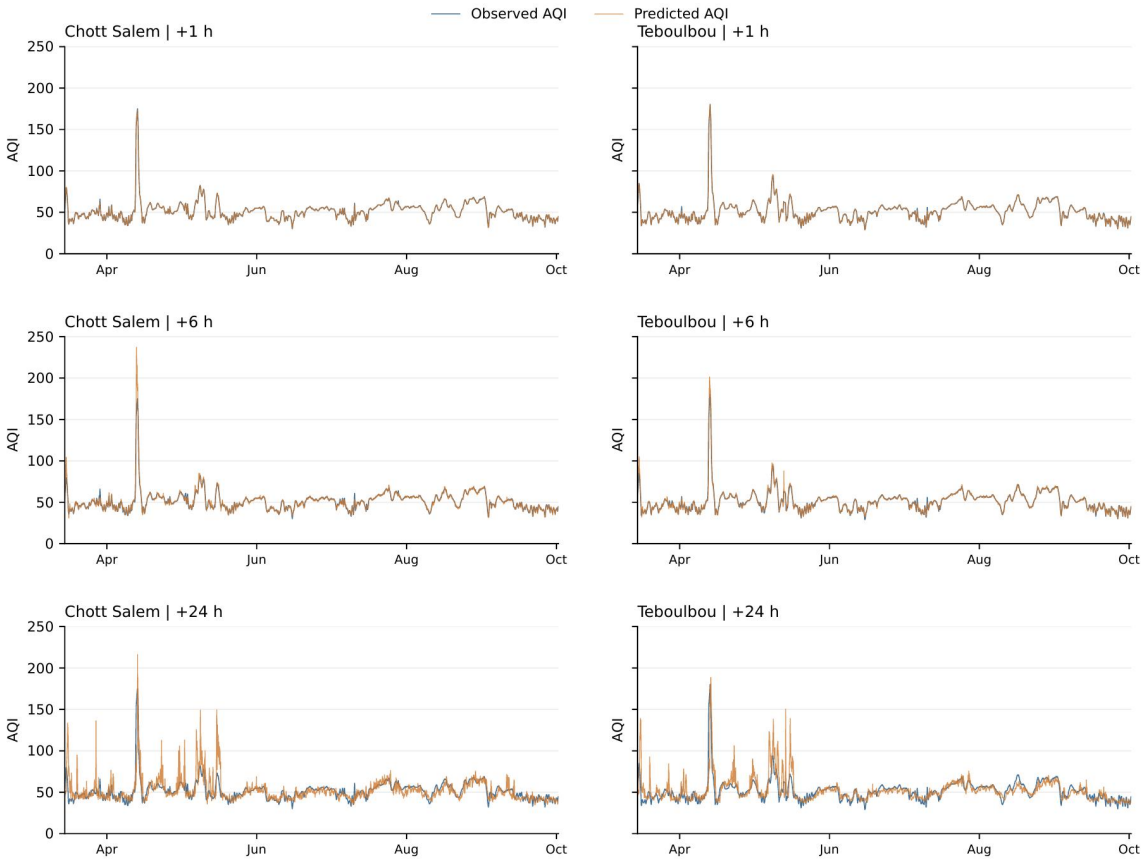



**Figure 9: Original full-period AQI curves for Chott Salem and Teboulbou, using the same horizons and axis scales as Figure 8. No smoothing or rescaling of predictions is introduced.**

The densely overlaid lines are useful for assessing seasonal coverage and broad departures, but cannot establish event-level accuracy by visual overlap alone. At short lead times, strong agreement is consistent with the information already supplied by current AQI and lagged predictors. At longer lead times, errors around abrupt transitions contribute more strongly to squared loss. Numerical errors and threshold diagnostics remain necessary because plotting thousands of points can conceal short-lived misses and compensating deviations.

The four-zone comparison also needs a spatial qualification. Similar patterns across panels may reflect shared meteorology, shared upstream grid products or geographic proximity. They are not, on their own, evidence that the model has learned independent neighbourhood dynamics. Quantifying grid-cell identity and cross-zone correlation would distinguish genuine spatial information from redundant input support.

## 5.4 Short-window inspection of neural forecasts

Figures 10 and 11 reproduce the final-seven-day inspection window from the archived prediction export. These are *DL-family* validation selections in Gabès Ville, not the global XGBoost selection used above. Keeping this distinction prevents an attractive local plot from being mistaken for the performance of the globally selected forecaster.

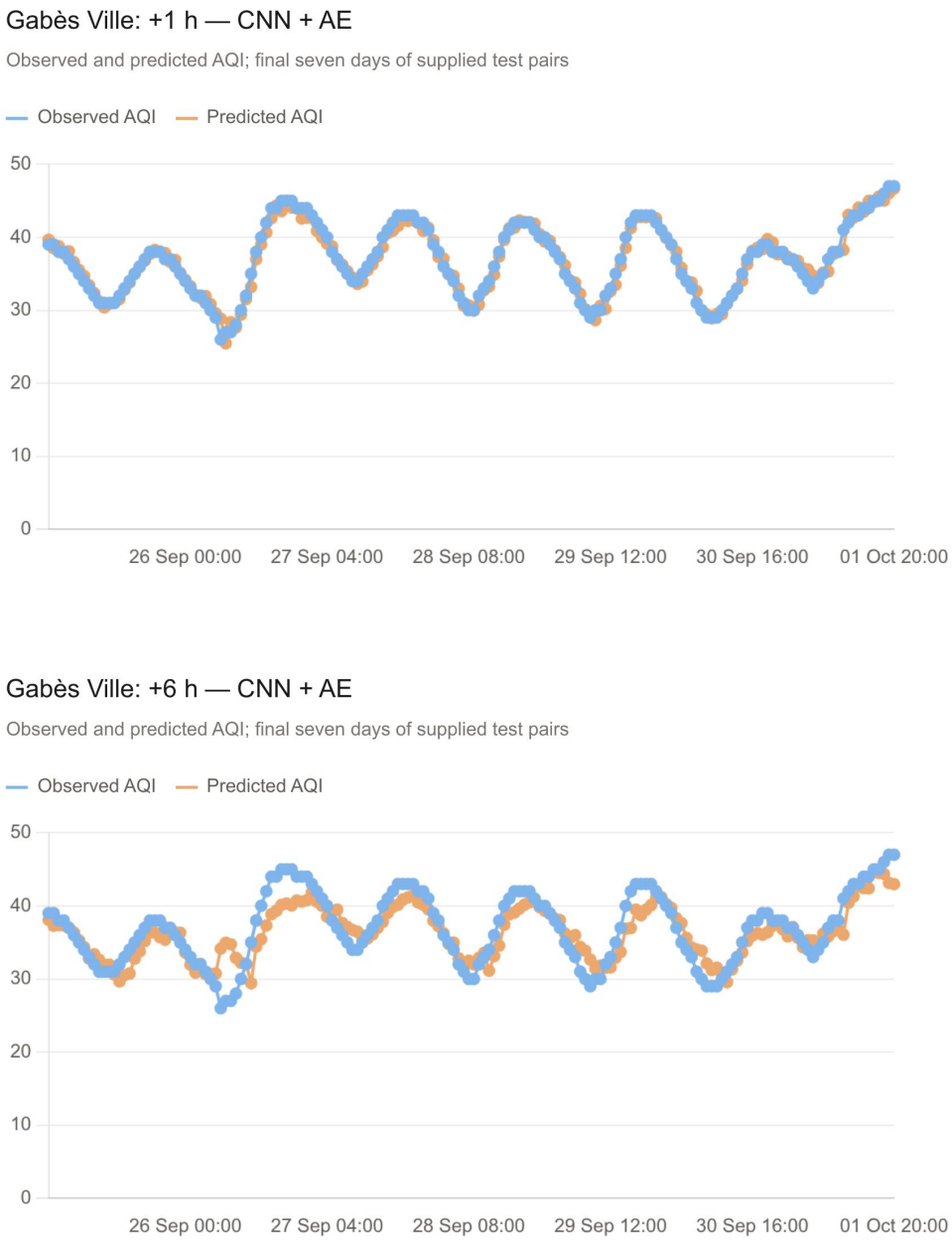



**Figure 10: Original final-seven-day CNN–AE forecasts in Gabès Ville: 1 hour above and 6 hours below. Selection is restricted to the DL family. Both panels retain the original common scale.**

The one-hour CNN–AE series closely tracks the visible oscillations, while the six-hour series displays greater departures around some changes in direction and amplitude. This is a qualitative reading of the original window, not an independently computed lag or peak-error statistic. The window ends with the test export and was not chosen to minimise visual disagreement; it must still remain secondary to full-period errors.

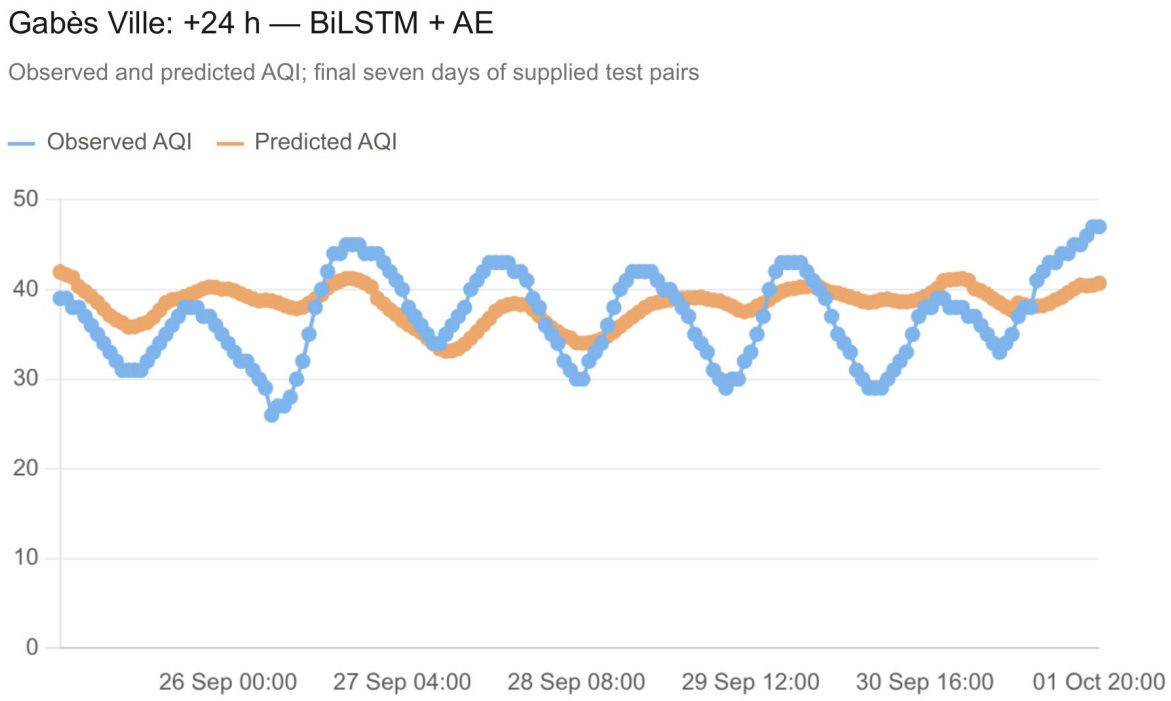



**Figure 11: Original day-ahead BiLSTM–AE forecast over the same Gabès Ville inspection window. The reduced forecast variation is retained as an observed limitation, without adjustment of the plotted values.**

The 24-hour BiLSTM–AE forecast has visibly lower variation than the reference in this window. It misses parts of the peak-to-trough range even when the mean level appears plausible. The original report gives a full-test RMSE of 10.910 AQI for this zone and model, compared with 4.772 over the displayed week. The smaller window error illustrates why visual examples cannot replace full-period performance. Compression towards common values is compatible with the plot, but whether it results from representation learning, loss weighting, optimisation or an unavailable driver requires additional experiments.

## 5.5 Recognition of high-AQI observations

Each horizon's global diagnostic contains 19,300 zone–time observations: 8,773 B1, 10,402 B2, 36 B3 and 89 B4. The highest two bands together constitute 125 observations, about 0.65% of the total. Shared and temporally clustered episodes mean that these are not necessarily 125 independent pollution events.

Figure 12 displays direct thresholding of the globally selected XGBoost forecasts. At 1 hour it correctly identifies 34 of 36 B3 and 88 of 89 B4 observations. At 6 hours the corresponding counts are 29 and 84. At 24 hours they fall to 14 and 33: recalls of 38.9% and 37.1%. Many B4 records are assigned to lower bands. Lower aggregate RMSE than persistence is therefore insufficient evidence of a dependable warning system.

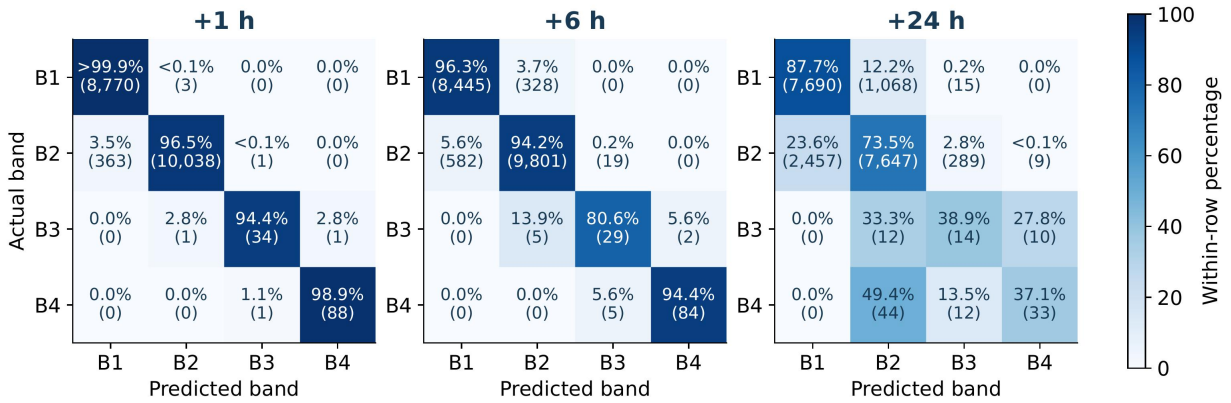



**Figure 12: Raw-threshold matrices for globally validation-selected XGBoost. Cells show row percentages and counts. B1: below 50; B2: 50–below 100; B3: 100–below 150; B4: at least 150 AQI.**

Stored classification summaries may use transformations that differ from direct regression thresholding. This revision does not merge those paths into one macro-F1 claim. Future reporting should retain the thresholds, class support, precision, recall and transformation for every output.

## 5.6 Evidence status of explanations and figures

The XAI-specific audit changes the earlier conclusion that the project had only unverified SHAP material. The implementation is real, is integrated with the model-training path, and produces stored outputs for all 84 units. The exact registry count is 288 rows: 24 global TreeSHAP, 24 beeswarm, 24 local TreeSHAP, 24 permutation, 24 PDP, 24 custom LIME-style, and 24 context/test-unit records for the 24 tabular units, plus 60 GradientExplainer and 60 temporal-importance records for the 60 neural units. The 24 context rows are the additional records omitted from the earlier arithmetic; they contain latest-test-prediction metadata rather than a seventh explanation algorithm. All exported rows are labelled as test-split records; the feature counts, sample counts and finite numerical payloads are internally consistent.

Figures 13–15 render stored XGBoost TreeSHAP values for Gabès Ville at the three horizons. Rendering uses the existing numerical exports; no model was loaded and no new attribution calculation was performed for this revision. The figures therefore demonstrate available, internally coherent XAI outputs, not a new experiment. The +1 h, +6 h and +24 h panels are descriptive unit-level inspections and are not evidence that a feature is a causal pollution source, that fuzzy context improves accuracy, or that explanations are invariant across zones and models.

The stored outputs also provide concrete descriptive findings beyond method presence. For the Gabès Ville XGBoost unit, global mean absolute TreeSHAP is led by `pm25_mean_24h` at +1 h (7.744) and +6 h (7.353), whereas +24 h is led by dust (5.525), pressure (5.144) and PM10 (4.272). The corresponding beeswarm payloads show signed, value-dependent behaviour: at +1 h, the upper quartile of `pm25_mean_24h` has mean SHAP +12.548 versus -9.751 for the lower quartile, and `aqi_current` has +3.871 versus -3.178. At +24 h, high dust values have mean SHAP +4.665 versus -6.113 for low values, while high pressure values have -5.323 versus +6.280 for low values. These are model-output associations in the stored test samples, not source-attribution claims.

Permutation importance gives a different, perturbation-based view. In the four XGBoost zone records, `pm25_lag_24` is the top stored RMSE-degradation feature at +6 h in all four zones; at +24 h, dew point is top in all four. The +1 h top feature varies across zones, with `pm25_lag_24` leading two records and dust lag or an AQI lag leading the others. The 63 stored XGBoost PDP curves are predominantly nondecreasing (38), with 19 non-monotone and 6 nonincreasing curves. For Gabès Ville, the +1 h PDP rises from 45.057 to 63.901 across the stored p5–p95 range of `pm25_mean_24h`;

at +24 h, the dust curve rises from 50.191 to 61.838 while the pressure curve falls from 59.227 to 49.448.

The custom *LIME-style local surrogate* records provide local rather than global evidence. Across the 12 XGBoost local records, the stored relative-MAE diagnostic ranges from 0.0515 to 0.0967, with median 0.0681, below the configured 0.20 screen. In the Gabès Ville examples, `pm25_mean_24h` is the largest positive local weight at +1 h and +6 h, while +24 h is led by PM2.5, dust and PM10, with pressure carrying a negative weight. These weighted-Ridge coefficients are not SHAP values and are not causal effects.

For the neural models, the 60 GradientExplainer summaries repeatedly place wind direction and hours since rain among the leading exported feature summaries: each appears in all 20 model–zone summaries at each horizon, although the exact ordering varies. The 60 temporal-importance records contain 24 steps for +1 h and +6 h and 168 steps for +24 h. Aggregating the stored values gives the largest mean temporal importance to the most recent lag (lag 1) at all three horizons; the result is a descriptive property of the archived sequence explanations and does not prove that recent observations cause forecast changes. These findings summarize existing numerical payloads only; no attribution was recalculated for this revision.

The implementation and output presence are verified, but the artifact records do not include immutable checkpoint hashes or a complete preprocessing-to-input manifest. Matching test metrics and checkpoint inventory support the same-run interpretation, yet the exact explanation-to-checkpoint link remains partial. The local TreeSHAP payloads also retain only leading contributions, preventing a full additivity check for most records. Accordingly, the article restores the XAI contribution in qualified form: the methods and outputs are reported, while causal, benchmark-wide and fully reproducible explanation claims are not made.

The other numerical result figures were checked against the archived prediction export: the RMSE/MAE comparison, aligned baselines, full-period curves, final-seven-day curves and raw-threshold matrices. The five architecture/system diagrams are methodological schematics; they describe the inspected implementation but do not constitute empirical performance evidence.

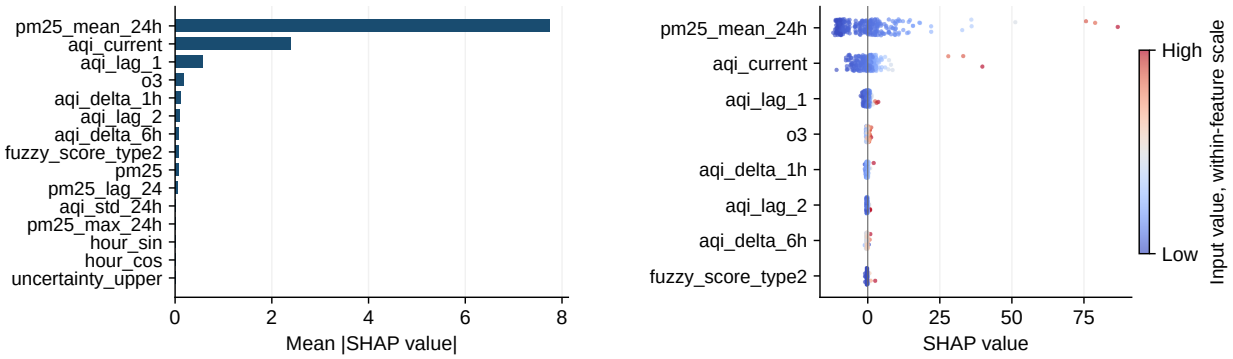



**Figure 13: Recorded XGBoost TreeSHAP for Gabès Ville at +1 h. The bar panel reports mean |SHAP| and the beeswarm reports signed SHAP values, both on the native model-output scale. Panels are rendered from stored global and beeswarm payloads (300 explained test observations); no new explanation was calculated. These are audit-linked descriptive outputs, not checkpoint-verified causal explanations.**

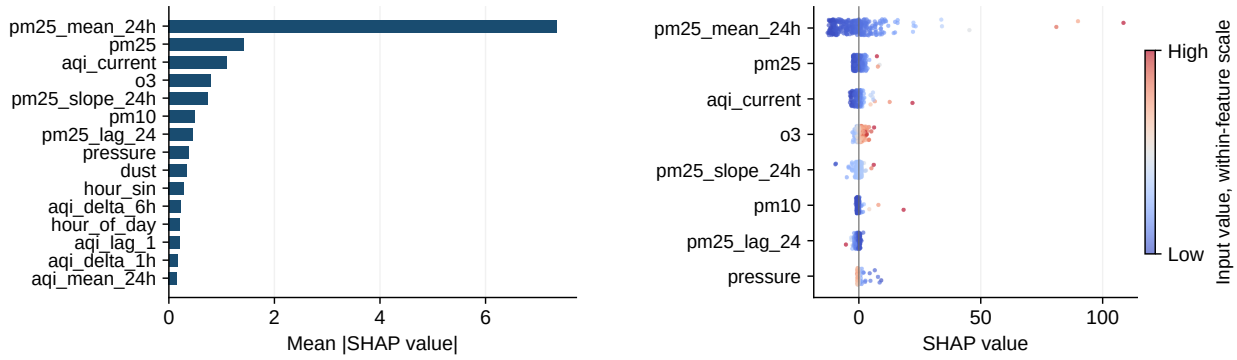



**Figure 14: Recorded XGBoost TreeSHAP for Gabès Ville at +6 h. The bar panel reports mean |SHAP| and the beeswarm reports signed SHAP values, both on the native model-output scale. The panel is descriptive and does not establish a horizon effect or causal feature importance.**

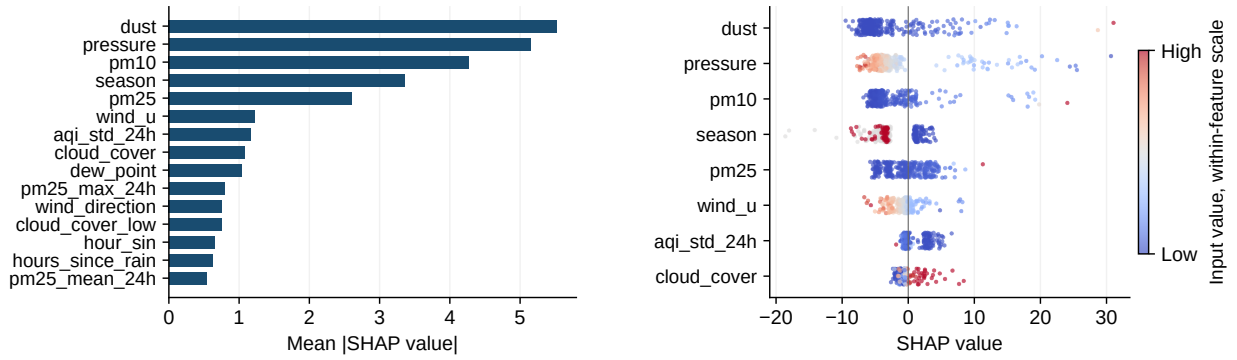



**Figure 15: Recorded XGBoost TreeSHAP for Gabès Ville at +24 h. The bar panel reports mean |SHAP| and the beeswarm reports signed SHAP values, both on the native model-output scale. The leading exported features differ from the shorter-horizon panels, but the result remains an audit-linked descriptive inspection rather than a causal or fully checkpoint-reproducible finding.**

## 5.7 Independent local-worker execution and supporting evidence

NAFASS separates forecasting tasks by zone, model and horizon. The Coordinator initializes a run manifest, creates filesystem-backed worker and task records, launches four local subprocesses concurrently, polls task and worker states, and records completion, failure and retry outcomes. Each of the four zone workers receives 21 model–horizon tasks, for 84 tasks in total; workers fit and evaluate their assigned tasks independently. The evidence consists of records such as `run.json`, `workers/*.json`, `tasks/*.json` and worker logs. This is independent zone-worker execution with local processes and filesystem-backed run evidence, not federated learning, synchronised gradients, multi-node execution or demonstrated speedup.

The separate local-worker run records completion of all 84 tasks after retries. Its test populations differ from those of the reference export; two zone–horizon groups also contain inconsistent fingerprints across models. Differences in those separate scores cannot be attributed to parallel execution. These records provide task-level provenance evidence, not speedup, federated learning, synchronised distributed training or equivalence of evaluated inputs. They belong in supplementary reproducibility material.

# 6 Discussion

## 6.1 Model capacity, information content and the role of simple references

The reported results favour a horizon-specific assessment rather than an unconditional preference for one AI family. XGBoost has strong short-horizon performance and remains below persistence in mean RMSE at all three horizons, whereas some recurrent models exceed even the simple reference by a wide margin. The scientific implication is that model complexity is not a substitute for informative inputs and reliable optimisation. Current AQI, engineered lags and trailing summaries give tree ensembles a strong representation before any recurrent state is learned.

This result should not be generalised into a claim that deep learning is unsuitable for air quality. Within this benchmark, CNN–AE and BiLSTM–AE differ substantially from plain recurrent configurations, and their ordering changes with lead time. Their losses, raw-input shortcuts and training schedules also differ. The comparison therefore does not isolate which component is responsible. A recurrent model with failed convergence, an incorrect inverse transform or a misaligned target could be penalised for implementation rather than representational limitations; these remain hypotheses requiring code and learning-curve inspection.

The validation–test ordering provides an additional lesson. Random Forest has the best descriptive six-hour test RMSE, but XGBoost is selected by validation. Reporting both is more defensible than retrospectively changing the selection. Whether this disagreement reflects seasonal distribution shift, finite validation support or tuning instability requires rolling-origin repetition. The present table does not establish a statistically significant advantage or a stable deployment ranking.

## 6.2 Relation to recent AI forecasting advances

Aurora demonstrates that large-scale pretraining can support different Earth-system forecasts, including air quality [19]. That finding motivates a transfer-learning comparator, not a claim that pretraining necessarily improves the Gabès task. A local AQI series, a globally gridded pollutant field and their evaluation units differ. The useful question is whether a pretrained model adds skill against matched local baselines when issue-time information, targets and compute budgets are controlled.

X2-AQFormer uses an explainable transformer for multi-day pollutant prediction [20]. It provides a methodological reference for testing feature-time attribution and multi-horizon modelling. NAFASS currently offers a narrower retrospective comparison with separate models and incomplete explanation-to-checkpoint linkage. Its next advance should therefore combine improved temporal validation with directly testable interpretability, rather than treat the presence of SHAP or attention as validation in itself.

Fahim and Karim's cross-city analysis explicitly considers foundation-model costs and covariate timing [22]. It illustrates why a forecast evaluated with information unavailable at issue time answers a different question from an operational forecast. For NAFASS, this distinction applies particularly to enriched historical weather, profile context and the precise upstream product. A competitive architecture cannot repair such a mismatch after evaluation.

The recent literature is therefore complementary to the present results rather than a league table. Our RMSE is measured in AQI units on a service-derived target; published particulate errors and cross-city scaled errors have different meanings. Neither smaller numeric error nor greater parameter count establishes superiority across those tasks. A future external dataset and common protocol would make direct methodological comparisons possible.

### 6.3 Rare episodes: why average error is insufficient

The 24-hour matrices reveal a practically consequential gap between aggregate regression skill and high-band recognition. B3 and B4 together represent only about 0.65% of evaluated zone–time observations. Their small support means that most training examples and most average error arise elsewhere. A forecaster can reduce the dominant contribution to RMSE while still missing a substantial fraction of the highest values.

The weekly day-ahead plot offers a compatible visual observation: a forecast can remain near the central level while underrepresenting amplitude. However, this neural example is not the same model as the globally selected XGBoost matrices. Their concurrence identifies an evaluation priority, not a shared failure mechanism proven across architectures. Episode-focused analysis should measure peak magnitude, timing, event recall and false alarms on a prespecified definition, retaining the dependence among successive hours and nearby zones.

SynCast explicitly studies particulate extremes and uses stochastic refinement in addition to deterministic forecasting [21]. Its relevance here is the separate evaluation of distributional tails; it does not establish that adding diffusion to this small benchmark would improve results. Before such an extension, simpler residual forecasting, validation-selected event weighting and calibrated exceedance probabilities should be compared under matched budgets. Forecast intervals should be assessed for empirical coverage specifically during high-index conditions, not only across common observations.

### 6.4 Interpretability, calibration and environmental plausibility

NAFASS does contain a genuine multi-method XAI implementation: TreeSHAP and beeswarms, permutation importance, PDPs, a custom LIME-style local surrogate, and GradientExplainer summaries for sequence models. The audit confirms stored outputs and internally consistent method metadata across the 84 units. This supports an auditable descriptive explanation layer and justifies retaining explainability in the article's scientific positioning. It does not establish a new XAI algorithm or make every archived explanation fully reproducible, because checkpoint hashes, exact input arrays and complete preprocessing manifests are not attached to the XAI rows.

The recorded Gabès Ville XGBoost panels show horizon-dependent attribution patterns, including strong contributions from recent AQI/particulate summaries at shorter leads and dust, pressure and PM10 among the leading exported features at +24 h. These are model associations on a service-derived AQI target. AQI is constructed from pollutant subindices; correlated predictors can substitute for one another; and native-output explanations may not describe a later calibration or classification transform. None of these results identifies an industrial source, proves atmospheric transport, or establishes toxicity or health benefit.

A stronger future analysis should attach checkpoint hashes, exact preprocessing and explained timestamps to every artifact, regenerate full local additivity records, compare attribution stability across repeated temporal fits, and evaluate grouped perturbations under physically plausible constraints. Attention remains an internal model quantity rather than a physical explanation. XAI complements, rather than replaces, forecasting metrics and rare-event diagnostics.

### 6.5 Environmental relevance and spatial scale in southern Tunisia

The North African context justifies a model attentive to dust and changing circulation, while the Gabès setting motivates local assessment near urban, industrial and coastal environments. The historical Tunisian and local deposition studies support that motivation [2, 3]. They do not provide station validation of the AQI target or identify the source of any particular predicted peak. Local environmental urgency should strengthen the demand for independent measurements rather than weaken it.

Service-derived products provide a consistent starting point, but may smooth local maxima and share information between nearby application coordinates. Apparent agreement across the four full-period figures can therefore reflect common upstream dynamics. Grid-cell identities, resolution and inter-zone redundancy must be verified before describing micro-scale spatial forecasting. Evaluation against independent stations would distinguish prediction of a provider series from prediction of observed air quality.

This distinction is also visible in recent AI research. SynCast separates reanalysis-based supervision from an observational assessment and discusses the limitations of gridded reference products [21]. An analogous separation is appropriate here. Current evidence supports retrospective forecasting of the archived series; it does not establish exposure reduction, regulatory compliance or a warning service with known event reliability.

### 6.6 From an application benchmark to a stronger AI contribution

Karmoude et al. identify generalisation and interpretability among the continuing challenges of AI-based air-quality research [23]. For this study, a more persuasive contribution requires a controlled experimental advance rather than a more elaborate name. One testable direction is horizon-dependent residual prediction relative to persistence, with simple and neural experts compared against the same origin-time information. The benefit of an adaptive ensemble would need comparison with single-model selection and fixed averaging; it is not an established result of this manuscript.

A compact new benchmark should retain persistence and seasonal persistence, a linear predictor, tree ensembles and carefully checked sequence models. Attention, reconstruction, the raw shortcut, fuzzy context and weighting should be assessed through matched ablations. Equal search budgets, several seeds and rolling-origin folds would help separate stable improvements from tuning variation. Multi-zone or graph learning is justified only if distinct spatial inputs and a credible spatial validation design support it.

Operational evaluation should additionally measure forecast-issue latency, availability of each predictor and repeated inference cost. Worker completion after retries supplies implementation evidence but cannot demonstrate speedup or statistical equivalence when test populations differ. The same principle applies to foundation models: their value should be evaluated under a clearly stated information and resource budget, not inferred from scale alone.

### 6.7 Limitations and reporting priorities

The strongest unresolved limitation is the absence of immutable linkage between the historical data, preprocessing revision and evaluated run. The corrected forward-fill loader cannot retroactively certify the old scores, and the historical lack of a verified horizon-width purge requires a controlled rerun. Acquisition payloads and operational release times must also be recovered. These issues affect validity, not merely documentation style.

The evaluation covers one region and one held-out period. Rolling-origin, seasonal and external tests remain needed, and nearby zones should not be treated as independent replicates. Uncertainty should account for temporal and spatial dependence; no confidence intervals or significance tests can be manufactured from the aggregate tables. The small, clustered high-band support further limits the precision of warning-oriented conclusions.

Finally, the profile-derived inputs require data-use and historical-version checks, and the XAI artifacts require checkpoint and input-manifest linkage. This revision renders three representative XAI figures from stored numerical exports but neither trains models nor creates new predictions or attribution values. The strengthened interpretation retains the implemented XAI contribution while limiting it to audit-linked descriptive behaviour. Its broader validity still requires the provenance

and rerun work described above.

## 7 Conclusion

This study reports an audited archived comparison of seven AI configurations for service-derived AQI forecasting in four zones of Gabès, southern Tunisia. The project implements a multi-method XAI layer, and its stored TreeSHAP, beeswarm, permutation, PDP, LIME-style and GradientExplainer outputs cover all 84 units. These outputs support audit-linked descriptive explanation, but incomplete checkpoint and input-manifest linkage prevents claims of fully reproducible or causal XAI findings. The coordinator–worker task records document local task execution; neither the XAI layer nor the worker layer establishes a new forecasting algorithm or an operational multi-node system. In the archived chronological benchmark, globally validation-selected XGBoost has lower mean RMSE than aligned persistence at 1, 6 and 24 hours. Relative improvement declines from about 51.7% to 13.9%, while day-ahead recognition of the two highest diagnostic bands remains limited. These findings favour horizon-specific assessment and explicit rare-observation diagnostics over a single aggregate accuracy claim.

The next scientific step is a controlled rerun authenticating upstream products, issue-time availability and preprocessing-to-checkpoint linkage. Matched ablations, modern baselines, rolling-origin and external evaluation, forecast uncertainty and station-based validation would strengthen the AI contribution. Current evidence concerns retrospective prediction of an archived service series, not a validated local warning system.

## Data, code and declarations

The reviewed materials contain an archived prediction export, model summaries, task records, and audit artifacts. No permanent public repository, executable environment, or complete immutable input-to-run manifest was verified for this revision. Data and code sharing therefore remain subject to confirmation of data-provider permissions and the authors' repository plan. The present manuscript does not claim full computational reproducibility from the archived export alone.

Author-specific statements on contributions, funding, competing interests, ethics, consent, and code availability must be completed from the authors' records before submission. The author names, affiliations, e-mail addresses and ORCID identifiers above must also be confirmed by all authors. If profile- or symptom-derived fields are retained in the input package, the applicable data-use, consent, and ethics statements must be supplied; aggregation alone does not establish exemption. No new model training, prediction, attribution calculation, or performance estimate was produced during this editorial correction.

## References


[1] World Health Organization. *WHO Global Air Quality Guidelines*. WHO, 2021. https://www.who.int/publications/i/item/9789240034228.

[2] Bouchlaghem K, Nsom B. Effect of atmospheric pollutants on the air quality in Tunisia. *The Scientific World Journal*, 2012:863528. https://doi.org/10.1100/2012/863528.

[3] Ahbil K, Sellami F, Baati H, Gautam S, Azri C. Influence of localized sources and meteorological conditions on dry-deposited particles: A case study of Gabès, Tunisia. *Science of the Total Environment*, 954:176726, 2024. https://doi.org/10.1016/j.scitotenv.2024.176726.

[4] Zheng Y, Yi X, Li M, et al. Forecasting fine-grained air quality based on big data. *KDD*, 2267–2276, 2015. https://doi.org/10.1145/2783258.2788573.

[5] Qi Z, Wang T, Song G, Hu W, Li X, Zhang Z. Deep Air Learning: Interpolation, Prediction, and Feature Analysis of Fine-Grained Air Quality. *IEEE Transactions on Knowledge and Data Engineering*, 30(12):2285–2297, 2018. https://doi.org/10.1109/TKDE.2018.2823740.

[6] Agbehadji IE, Obagbuwa IC. Systematic review of machine learning and deep learning techniques for spatiotemporal air quality prediction. *Atmosphere*, 15(11):1352, 2024. https://doi.org/10.3390/atmos15111352.

[7] Chen T, Guestrin C. XGBoost: A scalable tree boosting system. *KDD*, 785–794, 2016. https://doi.org/10.1145/2939672.2939785.

[8] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. https://arxiv.org/abs/1706.03762.

[9] Zeng A, Chen M, Zhang L, Xu Q. Are transformers effective for time series forecasting? Preprint arXiv:2205.13504, 2022. https://arxiv.org/abs/2205.13504.

[10] Bai S, Kolter JZ, Koltun V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. Preprint arXiv:1803.01271, 2018. https://arxiv.org/abs/1803.01271.

[11] Nie Y, Nguyen NH, Sinthong P, Kalagnanam J. A time series is worth 64 words: Long-term forecasting with transformers. Preprint arXiv:2211.14730, revised 2023. https://arxiv.org/abs/2211.14730.

[12] Roberts DR, Bahn V, Ciuti S, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. https://doi.org/10.1111/ecog.02881.

[13] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. https://doi.org/10.1016/j.patter.2023.100804.

[14] Lundberg SM, Lee SI. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 2017. https://arxiv.org/abs/1705.07874.

[15] Ribeiro MT, Singh S, Guestrin C. Why should I trust you? Explaining the predictions of any classifier. *KDD*, 1135–1144, 2016. https://doi.org/10.1145/2939672.2939778.

[16] Open-Meteo. Air Quality API documentation. https://open-meteo.com/en/docs/air-quality-api.

[17] Inness A, Ades M, Agustí-Panareda A, et al. The CAMS reanalysis of atmospheric composition. *Atmospheric Chemistry and Physics*, 19:3515–3556, 2019. https://doi.org/10.5194/acp-19-3515-2019.

[18] Akiba T, Sano S, Yanase T, Ohta T, Koyama M. Optuna: A next-generation hyperparameter optimization framework. *KDD*, 2623–2631, 2019. https://doi.org/10.1145/3292500.3330701.

[19] Bodnar C, Bruinsma WP, Lucic A, et al. A foundation model for the Earth system. *Nature*, 641:1180–1187, 2025. https://doi.org/10.1038/s41586-025-09005-y.

[20] Zhang Z, Schlesinger D, Johansson C, Engardt M, Ma X. X2-AQFormer: unveiling dynamic drivers in multi-day hourly air pollution forecasting. *npj Clean Air*, 2:17, 2026. https://doi.org/10.1038/s44407-026-00058-5.

[21] Abeysinghe Y, Munir MA, Baliah S, Sarafian R, Khan FS, Rudich Y, Khan S. Synergistic neural forecasting of air pollution with stochastic sampling. *npj Clean Air*, 2:36, 2026. https://doi.org/10.1038/s44407-026-00076-3.

[22] Fahim MMM, Karim MR. Cost-aware evaluation of time-series foundation models for urban air-quality and temperature forecasting. *Scientific Reports*, 2026. https://doi.org/10.1038/s41598-026-73372-3.

[23] Karmoude M, Munhungewarwa B, Chiraira I, et al. Machine learning for air quality prediction and data analysis: Review on recent advancements, challenges, and outlooks. *Science of the Total Environment*, 1002:180593, 2025. https://doi.org/10.1016/j.scitotenv.2025.180593.